The Three-Generation Problem in the Metric Bundle Framework

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Abstract

We investigate the three-generation problem within the metric bundle framework, where the Pati-Salam gauge group $\mathrm{SU}(4) \times \mathrm{SU}(2)_L \times \mathrm{SU}(2)_R$ and one generation of fermions emerge from $Y^{14} = \mathrm{Met}(X^4)$. The algebraic construction yields exactly one **16**-plet of Spin(10); the observed three generations require a topological or geometric mechanism. We examine five candidate explanations: (i) the index theorem on Y^{14}, constrained by the contractibility of the fibre; (ii) a quaternionic structure on the positive-norm sector, providing a natural multiplicity of three from the three independent complex structures of $\mathbb{H}$; (iii) topological sectors of the base spacetime X^4 via $\pi_3(\mathrm{SO}(4)) \cong \mathbb{Z} \oplus \mathbb{Z}$; (iv) a $\mathbb{Z}_3$ discrete symmetry from the cubic structure of the Pati-Salam Yukawa couplings; and (v) the gravitational anomaly constraint $16\, N_G \equiv 0 \pmod{24}$, which selects $N_G \in \{3, 6, 9, \ldots\}$. While none of these mechanisms provides a complete derivation of $N_G = 3$, the quaternionic route and the anomaly constraint together strongly suggest three as the preferred value. We formulate this as a precise conjecture and identify the calculations needed to settle it.

Keywords: three generations, metric bundle, quaternionic structure, Atiyah-Singer index, Pati-Salam model

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1 Introduction

The metric bundle programme [1–4] derives the Pati-Salam gauge group, one generation of fermions, and the correct signs for gravity and gauge fields from the geometry of $Y^{14} = \mathrm{Met}(X^4)$. However, a glaring gap remains: the algebraic construction produces exactly *one* copy of the **16**-plet of Spin(10), while nature requires *three* generations of fermions.

The three-generation problem is one of the deepest unsolved problems in particle physics. No known theoretical framework—not the Standard Model, not string theory (without specific compactification choices), not loop quantum gravity—provides a first-principles derivation of $N_G = 3$. In GUT models, N_G is typically determined by the topology of the internal space (e.g., the Euler characteristic of a Calabi-Yau manifold in string theory [5]).

In this paper, we investigate whether the metric bundle framework contains a natural mechanism for $N_G = 3$. We examine five approaches, ranging from the well-established (index theorem) to the speculative (quaternionic structure). Our goal is not to claim a solution but to map the landscape of possibilities and identify the most promising route.

Outline. Section 2 reviews why the algebraic construction gives one generation. Section 3 examines the index theorem approach. Section 4 explores the quaternionic structure. Section 5 considers topological sectors. Section 6 discusses discrete symmetries. Section 7 revisits the anomaly constraint. Section 8 formulates a precise conjecture and programme.

2 One Generation from Algebra

The fermion content of the metric bundle theory is derived in [2] from the Clifford algebra of the positive-norm sector of the normal bundle:

$$\mathrm{Cl}(\mathbb{R}^6, g_+) \otimes_{\mathbb{R}} \mathbb{C} \;\cong\; M_8(\mathbb{C})\,, \tag{1}$$

acting on the spinor $\mathbb{C}^8 = S^+ \oplus S^- = \mathbf{4} \oplus \bar{\mathbf{4}}$ of $\mathrm{Spin}(6) \cong \mathrm{SU}(4)$. Combined with $\mathrm{Spin}(4) \cong \mathrm{SU}(2)_L \times \mathrm{SU}(2)_R$ from the negative-norm sector, the total spinor is:

$$\mathbf{16} = (\mathbf{4}, \mathbf{2}, \mathbf{1}) \oplus (\bar{\mathbf{4}}, \mathbf{1}, \mathbf{2})\,. \tag{2}$$

This gives exactly *one* **16**-plet. The Clifford algebra construction is algebraic and produces a single irreducible spinor module. To obtain multiple generations, one needs either:

1. A *topological* mechanism (index theorem, topological sectors),
2. An *algebraic* enhancement (larger Clifford algebra, division algebra structure), or
3. A *geometric* multiplicity (multiple complex structures, moduli).

3 The Index Theorem Approach

3.1 General framework

In Kaluza-Klein theories, the number of chiral zero modes of the Dirac operator on the total space is given by the Atiyah-Singer index theorem:

$$N_G = \mathrm{ind}(\not{D}) = \int_Y \hat{A}(TY) \wedge \mathrm{ch}(V)\,, \tag{3}$$

where $\hat{A}$ is the $\hat{A}$-genus and $\mathrm{ch}(V)$ is the Chern character of the gauge bundle.

For a fibre bundle $\pi\colon Y \to X$ with fibre F, the Atiyah-Singer family index theorem [6] gives:

$$\mathrm{ind}(\not{D}_Y) = \int_X \hat{A}(TX) \wedge \mathrm{ch}(\mathrm{ind}(\not{D}_F))\ , \tag{4}$$

where $\mathrm{ind}(\not{D}_F)$ is the index bundle of the family of Dirac operators on the fibres.

3.2 Application to the metric bundle

For the metric bundle $Y^{14} = \mathrm{Met}(X^4)$ with fibre $F = \mathrm{GL}^+(4,\mathbb{R})/\,\mathrm{SO}(4)$:

Proposition 3.1 (Fibre index obstruction)**.** *The fibre $F = \mathrm{GL}^+(4,\mathbb{R})/\,\mathrm{SO}(4) \cong \mathbb{R}^{10}$ is contractible. Therefore:*

1. *All characteristic classes of F vanish: $\hat{A}(TF) = 1$, $p_i(TF) = 0$.*
2. *The Euler characteristic $\chi(F) = 1$.*
3. *The index of the Dirac operator on F (with appropriate boundary conditions) is determined by the structure at infinity, not by the topology of F.*

Proof. $F = \mathrm{GL}^+(4,\mathbb{R})/\,\mathrm{SO}(4)$ is diffeomorphic to $\mathbb{R}^{>0} \times S^2_0(\mathbb{R}^4) \cong \mathbb{R}^{10}$ (the positive reals times the traceless symmetric matrices), which is contractible. All homotopy groups vanish: $\pi_n(F) = 0$ for all n. □

Remark 3.2. The contractibility of the fibre is a significant obstruction to the index theorem approach. In string compactifications, the internal space has non-trivial topology (e.g., a Calabi-Yau threefold with $\chi/2 = 3$). The metric bundle fibre has trivial topology.

3.3 Non-compact fibre and L^2 index theory

For non-compact fibres, the relevant index theorem is the L^2-index theorem [7], which counts normalisable zero modes. On the non-compact fibre $F = \mathrm{GL}^+(4,\mathbb{R})/\,\mathrm{SO}(4)$, the number of L^2-normalisable harmonic spinors depends on the asymptotic behaviour of the metric.

For the DeWitt metric, the fibre has finite-volume cusps in the conformal direction (the negative-norm direction). This is analogous to the situation for locally symmetric spaces, where the L^2-index is related to the volume of the fundamental domain and the Euler characteristic of the Borel-Serre compactification.

Conjecture 3.3 (Fibre L^2-index)**.** *The L^2-index of the Dirac operator on $F = \mathrm{GL}^+(4,\mathbb{R})/\,\mathrm{SO}(4)$ with the DeWitt metric, twisted by the appropriate gauge bundle, is zero. Any non-trivial generation count must arise from the* base *topology or the* twisting *of the fibre bundle, not from the fibre itself.*

3.4 Base topology contribution

If the fibre contributes trivially to the index, the generation number is determined by the base topology via (4). For a compact base X^4 with spin structure:

$$\mathrm{ind}(\not{D}_X) = \int_{X^4} \hat{A}(TX) = -\frac{1}{24}\int_{X^4} p_1(TX) = -\frac{\sigma(X)}{8}\,, \tag{5}$$

where $\sigma(X)$ is the signature of X. For $N_G = 3$, we would need $\sigma(X) = -24$ (or some appropriate modification involving the gauge bundle).

This is a strong topological constraint on spacetime. It is not clear whether the metric bundle framework selects a specific base topology, but the requirement $\sigma(X) \sim -24$ is intriguing in light of the role of 24 in the theory of modular forms and the $\hat{A}$-genus.

4 The Quaternionic Structure

4.1 Three complex structures from the quaternions

The most natural candidate for a "threeness" in the metric bundle is the quaternionic structure of the positive-norm sector.

Proposition 4.1 (Three complex structures on $\mathbb{R}^6$)**.** *The six-dimensional positive-norm sector* $(\mathbb{R}^6, g_+)$ *of the DeWitt metric admits three linearly independent complex structures* I, J, K *(each satisfying* $I^2 = J^2 = K^2 = -\mathbf{1}$*), arising from the quaternionic structure of* $\mathbb{R}^4$*.*

Proof. The space $\mathbb{R}^6$ carries the representation of $\mathrm{SO}(6) \cong \mathrm{SU}(4)/\mathbb{Z}_2$. Under the inclusion $\mathrm{SU}(2) \hookrightarrow \mathrm{SU}(4)$ (via the identification $\mathrm{SU}(4) \supset \mathrm{Sp}(2) \supset \mathrm{SU}(2)$), the fundamental $\mathbf{4}$ restricts to $\mathbf{2} \oplus \mathbf{2}$, and the $\mathbf{6}$ of $\mathrm{SO}(6)$ (the antisymmetric $\Lambda^2(\mathbf{4})$) restricts to $\mathbf{3} \oplus \mathbf{3}$.

More concretely: the Weyl tensor sector $W^+ \oplus W^-$ (three self-dual plus three anti-self-dual components) has a natural identification $W^+ \cong \mathbb{R}^3$, $W^- \cong \mathbb{R}^3$. The three linearly independent complex structures on $\mathbb{R}^4 \cong \mathbb{H}$ (from left multiplication by i, j, k) induce, via the functorial action on $\Lambda^2(\mathbb{R}^4)$, three complex structures on $\mathbb{R}^6 = \Lambda^2_+(\mathbb{R}^4) \oplus \Lambda^2_-(\mathbb{R}^4)$. Each is an endomorphism $J_a \in (\mathbb{R}^6)$ with $J_a^2 = -\mathbf{1}$, since 6 is even.

Note: Since $\dim \mathbb{R}^6 = 6$ is not divisible by 4, the triple (I, J, K) does not satisfy the full quaternion algebra $IJK = -\mathbf{1}$ as endomorphisms of $\mathbb{R}^6$. However, the three complex structures are inequivalent (their centralisers in $\mathfrak{so}(6)$ give three distinct $\mathrm{U}(3)$ subalgebras), and each independently defines a breaking $\mathrm{SO}(6) \to \mathrm{U}(3) \to \mathrm{SU}(3) \times \mathrm{U}(1)$. □

4.2 Three generations from three complex structures

In [2], the fermion content is derived by choosing a *single* complex structure J on $\mathbb{R}^6$, which breaks $\mathrm{SO}(6) \to \mathrm{U}(3) \to \mathrm{SU}(3) \times \mathrm{U}(1)_{B-L}$. The choice of J is not unique: the quaternionic structure provides three independent choices I, J, K.

Conjecture 4.2 (Three generations from $\mathbb{H}$)**.** *The three generations of fermions correspond to the three inequivalent complex structures* (I, J, K) *on the positive-norm sector* $\mathbb{R}^6$*. Each complex structure defines an independent* $\mathrm{SU}(3) \times \mathrm{U}(1)$ *breaking and hence an independent set of fermion quantum numbers. The three sets are related by the* $\mathrm{SU}(2)_R$ *"flavour" symmetry that rotates* (I, J, K)*.*

Remark 4.3. This conjecture connects to several known approaches:

1. **Division algebras:** The division algebra approach of Furey [8] and Dixon [9] uses $\mathbb{C}\otimes\mathbb{H}\otimes\mathbb{O}$ to obtain three generations. The $\mathbb{H}$ factor provides the "threeness" via the three imaginary quaternions. Our conjecture is analogous: $\mathbb{H}$ acts on the normal bundle via the quaternionic structure.
2. **Atiyah's conjecture:** Atiyah conjectured that the number of particle generations is related to the quaternionic structure of spacetime [10]. In the metric bundle, the quaternionic structure appears naturally on the positive-norm sector.
3. **Twistor theory:** The three complex structures on $\mathbb{R}^6$ are the analogue of the sphere $S^2 \cong \mathbb{C}P^1$ of complex structures on $\mathbb{R}^4$ in twistor theory. The parameter space is $\mathrm{SO}(6)/\,\mathrm{U}(3) \cong \mathbb{C}P^3$ (the twistor space of $\mathbb{R}^6$), and the three quaternionic complex structures are three special points in this space.

4.3 Obstruction: independence of generations

A key question for Conjecture 4.2 is whether the three complex structures produce genuinely *independent* fermion species or merely relabel the same fermion.

Proposition 4.4 (Non-equivalence of complex structures)**.** *The three complex structures I, J, K on $\mathbb{R}^6$ define three* inequivalent $\mathrm{SU}(3)\times\mathrm{U}(1)$ *subalgebras of* $\mathrm{SU}(4)$*. Specifically, the centralisers satisfy:*

$$Z_{\mathrm{SU}(4)}(I) \neq Z_{\mathrm{SU}(4)}(J) \neq Z_{\mathrm{SU}(4)}(K)\,, \tag{6}$$

and the pairwise intersections are:

$$Z_{\mathrm{SU}(4)}(I) \cap Z_{\mathrm{SU}(4)}(J) = Z_{\mathrm{SU}(4)}(\{I,J,K\}) = \mathrm{Sp}(2)\cap\mathrm{SU}(4)\,. \tag{7}$$

This means the three breaking patterns $\mathrm{SU}(4) \to \mathrm{SU}(3)_i \times \mathrm{U}(1)_i$ for $i = I, J, K$ produce three *different* $\mathrm{SU}(3)$ subgroups. The fermion representations under these three $\mathrm{SU}(3)$s are unitarily inequivalent, so they describe genuinely distinct fermion species.

4.4 The flavour $\mathrm{SU}(2)$

The quaternionic structure carries an $\mathrm{SU}(2)$ symmetry: the group that rotates (I, J, K) by the adjoint action of $\mathbb{H}^*$ on $\mathrm{Im}(\mathbb{H}) \cong \mathbb{R}^3$. In the division algebra language, this is the "family symmetry" $\mathrm{SU}(2)_F$.

Remark 4.5. If $\mathrm{SU}(2)_F$ is a gauge symmetry, it would predict inter-generation gauge bosons ("familons"), which are not observed. More likely, $\mathrm{SU}(2)_F$ is *broken* by the choice of vacuum (the specific metric section $g\colon X \to Y$), leaving only a residual discrete symmetry (e.g., $\mathbb{Z}_3 \subset \mathrm{SU}(2)_F$). The breaking pattern would determine the fermion mass hierarchy and mixing angles.

5 Topological Sectors

5.1 Homotopy of the structure group

The structure group of the metric bundle is $\mathrm{GL}^+(4,\mathbb{R})$, whose homotopy groups include:

$$\pi_0(\mathrm{GL}^+(4,\mathbb{R})) = 0\,, \quad \pi_1(\mathrm{GL}^+(4,\mathbb{R})) = \mathbb{Z}_2\,, \quad \pi_3(\mathrm{GL}^+(4,\mathbb{R})) \cong \pi_3(\mathrm{SO}(4)) \cong \mathbb{Z}\oplus\mathbb{Z}\,. \tag{8}$$

The last group is relevant for four-dimensional physics: instantons are classified by $\pi_3(G)$, and $\pi_3(\mathrm{SO}(4)) \cong \mathbb{Z} \oplus \mathbb{Z}$ has *two* independent instanton numbers.

5.2 Double instanton structure

The isomorphism $\mathrm{SO}(4) \cong [\mathrm{SU}(2)_L \times \mathrm{SU}(2)_R]/\mathbb{Z}_2$ gives $\pi_3(\mathrm{SO}(4)) \cong \pi_3(\mathrm{SU}(2)_L) \oplus \pi_3(\mathrm{SU}(2)_R) \cong \mathbb{Z} \oplus \mathbb{Z}$. An instanton is characterised by two integers (n_L, n_R):

$$n_L = \frac{1}{8\pi^2} \int_{S^4} \mathrm{tr}(F_L \wedge F_L)\,, \qquad n_R = \frac{1}{8\pi^2} \int_{S^4} \mathrm{tr}(F_R \wedge F_R)\,. \tag{9}$$

Conjecture 5.1 (Generation number from instantons)**.** *The number of fermion generations is related to the instanton content of the vacuum. In the metric bundle, the three generations may correspond to three topologically distinct sectors characterised by specific values of* (n_L, n_R)*.*

This is highly speculative. The standard instanton mechanism produces fermion zero modes (via the Atiyah-Singer index theorem), but the number of zero modes depends on the specific instanton configuration, not on a fixed topological invariant of the internal space.

5.3 π_7 and the exceptional periodicity

At a deeper level, the generation number might be related to higher homotopy groups. The relevant group for the metric bundle is:

$$\pi_7(\mathrm{SO}(6)) \cong \mathbb{Z}\,, \tag{10}$$

which classifies topologically non-trivial gauge field configurations in the SU(4) sector over S^8 (or its dimensional reduction). The connection to three generations is unclear but deserves investigation.

6 Discrete Symmetries

6.1 $\mathbb{Z}_3$ from the Yukawa structure

The Pati-Salam Yukawa coupling involves the **16**-plet, the Higgs $(1, 2, 2)$, and a second **16**-plet. The gauge-invariant Yukawa term is:

$$\mathcal{L}_Y = y\, \epsilon^{abcd}\, \psi_L^{a\alpha}\, \phi_{\alpha\dot{\beta}}\, \psi_R^{b\dot{\beta}}\, \delta_{cd} + \text{h.c.} \tag{11}$$

The SU(4) contraction $\epsilon^{abcd}\delta_{cd}$ involves the antisymmetric tensor, which has a $\mathbb{Z}_3$ symmetry under cyclic permutations of the first three indices (a, b, c) with d fixed.

Remark 6.1. This $\mathbb{Z}_3$ acts on the SU(4) indices, not on the generation indices. It does not directly explain three generations. However, in models where the SU(4) colour indices and generation indices are related (e.g., via a "colour-generation locking"), such a $\mathbb{Z}_3$ could play a role.

7 The Anomaly Constraint Revisited

As shown in [4], the gravitational anomaly constraint requires:

$$16\, N_G \equiv 0 \pmod{24}\,, \tag{12}$$

which gives $N_G \in \{3, 6, 9, \ldots\}$ (i.e., N_G must be a multiple of 3).

Combined with phenomenological constraints:

1. $N_G \geq 3$ for the CKM phase (CP violation) [11].
2. $N_G \leq 8$ from the requirement that $\mathrm{SU}(3)_c$ remain asymptotically free at one loop [12]: the beta function coefficient $b_0 = 11 - \frac{4}{3}N_G > 0$ requires $N_G \leq 8$.
3. $N_G \leq 3$ from precision measurements of the Z boson width (for light neutrinos) [13].

The intersection of these constraints is $N_G = 3$.

Remark 7.1. The gravitational anomaly constraint is necessary but not sufficient. It permits $N_G = 3$ but also $N_G = 6, 9, \ldots$ The phenomenological constraints (especially the Z width) are external inputs, not derived from the framework. A complete solution would derive $N_G = 3$ from the geometry of the metric bundle alone.

8 A Conjecture and Programme

8.1 The main conjecture

Based on the analysis above, we formulate:

Conjecture 8.1 (Three generations)**.** *The number of fermion generations in the metric bundle framework is $N_G = 3$, arising from the quaternionic structure of the positive-norm sector $(\mathbb{R}^6, g_+)$ of the DeWitt metric. Specifically:*

1. *The three complex structures (I, J, K) on $\mathbb{R}^6$ define three inequivalent $\mathrm{SU}(3) \times \mathrm{U}(1)$ subalgebras of $\mathrm{SU}(4)$.*
2. *Each complex structure produces an independent set of fermion zero modes via the Dirac operator on Y^{14} twisted by the corresponding gauge bundle.*
3. *The resulting three* **16***-plets transform as a triplet of the "family $\mathrm{SU}(2)_F$" that rotates (I, J, K).*
4. *The family $\mathrm{SU}(2)_F$ is broken by the vacuum, giving mass splittings and mixing angles.*

8.2 Required calculations

To establish or refute Conjecture 8.1, the following calculations are needed:

1. **Quaternionic complex structures:** Explicitly construct the three complex structures (I, J, K) on the six-dimensional positive-norm sector in terms of the DeWitt basis. Verify that they satisfy $I^2 = J^2 = K^2 = IJK = -\mathbf{1}$ and that they commute with the $\mathrm{SU}(4)$ action in the appropriate sense.

2. **Dirac index with twist:** Compute the L^2-index of the Dirac operator on Y^{14} twisted by the gauge bundles associated to each complex structure. This requires extending the index theorem to the non-compact setting of the metric bundle.

3. **Zero mode counting:** For each complex structure, determine the space of harmonic spinors (zero modes of the twisted Dirac operator) on the fibre. If each complex structure produces exactly one zero mode, and the zero modes are linearly independent, then $N_G = 3$.

4. **Family symmetry breaking:** Determine how the family $\mathrm{SU}(2)_F$ is broken by the vacuum metric $\bar{g}$. Compute the resulting Yukawa matrix structure and check whether it is compatible with the observed mass hierarchy and CKM/PMNS mixing.

5. **Stability:** Verify that the three-generation configuration is dynamically stable, i.e., that it minimises the effective potential.

8.3 Alternative: K-theory and the fibre

An alternative approach uses K-theory to classify the possible fermion sectors. The relevant K-group is:

$$K^0(F) = K^0(\mathrm{GL}^+(4,\mathbb{R})/\,\mathrm{SO}(4)) = K^0(\mathbb{R}^{10}) = \mathbb{Z}\,, \tag{13}$$

which classifies vector bundles over the fibre. Since $K^0(F) = \mathbb{Z}$, there is a single "charge" that counts the net number of chiral fermions. The value of this charge is determined by the specific gauge bundle.

For the base, the relevant K-group is:

$$K^0(X^4) = \mathbb{Z} \oplus \tilde{K}^0(X^4)\,, \tag{14}$$

where $\tilde{K}^0(X^4)$ depends on the topology of spacetime. The generation number may be related to $\tilde{K}^0(X^4)$ via the Thom isomorphism in K-theory.

9 Discussion

We have examined five approaches to the three-generation problem in the metric bundle framework:

Approach	**Gives $N_G = 3$?**	**Status**
Index theorem on Y^{14}	Obstructed by contractible fibre	Unlikely
Quaternionic structure	3 complex structures on $\mathbb{R}^6$	Promising
Topological sectors	Two instanton numbers (n_L, n_R)	Speculative
Discrete $\mathbb{Z}_3$ symmetry	$\mathbb{Z}_3$ in Yukawa structure	Insufficient
Anomaly constraint	$N_G \equiv 0 \pmod 3$	Necessary, not sufficient

The most promising route is the *quaternionic structure* (Conjecture 8.1), which provides a natural "threeness" from the three imaginary quaternions. Combined with the

anomaly constraint (N_G must be a multiple of 3), this gives $N_G = 3$ as the unique minimal solution.

The quaternionic route has the additional appeal of connecting to the division algebra programme [8, 9], where $\mathbb{C} \otimes \mathbb{H} \otimes \mathbb{O}$ provides three generations via the three independent complex structures within $\mathbb{H}$. In the metric bundle, the quaternionic structure arises *geometrically* from the positive-norm sector rather than being postulated algebraically.

What would settle the question. The key calculation is the Dirac index on Y^{14} with the three different gauge bundle twists corresponding to (I, J, K). If each twist produces exactly one zero mode, the three-generation problem is solved. If not, the quaternionic mechanism fails and one must look elsewhere.

A go/no-go criterion. As a first check: if the three complex structures (I, J, K) produce three linearly independent elements of $H^0(Y, S(V))$ (the kernel of the Dirac operator on spinors twisted by V), the mechanism works. If they produce the *same* element (related by the $\mathrm{SU}(2)_F$ rotation), then the quaternionic structure gives multiplicity one, not three, and the mechanism fails.

This check requires only local (fibre) information and can be performed computationally using the explicit DeWitt metric and complex structures from [2]. We leave this calculation for future work.

Connection to quantum observation. Paper VI [14] derives the Born rule and uncertainty relations from finite Markov blanket observation of the $(6, 4)$ DeWitt geometry. The mutually unbiased bases (MUBs) that underpin that derivation form a natural structure on the quaternionic sector; whether this selects the three complex structures (I, J, K) as geometrically distinguished is an open question that may connect the generation problem to quantum observation geometry.

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